Leaders

Components, Constraints, And Market Demands: NAVIGATING THE RF ECOSYSTEM

Drawing on experience spanning satellite systems, connectivity and RF engineering, CEO Jin Bains outlines how Mini-Circuits aligns technical constraints with high-volume RF market demands.



JIN BAINS
CEO, MINI-CIRCUITS

Your career spans test and measurement, large-scale connectivity, and satellite systems. How has that shaped your perspective as a components company CEO?

It broadened my horizons in a big way. Hewlett-Packard (HP) was a great place to learn RF and microwave, and as an early engineer, it was a fantastic place to be. I developed through that experience, moving into management and leadership positions before joining National Instruments to build up the RF and wireless division there. At Facebook, I served as the head of Southern California Connectivity, which involved finding innovative ways to bring internet access to the world. I was able to think broadly about novel technologies, whether in Wi-Fi, cellular, or space and satellite systems. There was a lot of space-satellite work in that timeframe, and it allowed me to go to countries like Kenya, Peru, and other places in South America, where there was less connectivity. We were looking for ways to get cost-effective internet connectivity to those areas. That experience at Facebook gave me a broad sense of what was happening in the world. Then I had the opportunity to join the Amazon program, which was very much a mega-constellation effort. Again, it was global, a large investment, highly challenging, and very thrilling to be part of.

What did that 'system-level' experience change in how you think about components?

It really reinforced the move into space. We did extensive work around cost-effective satellites with high data rates, connecting them together through inter-satellite links using optical technology. I spent a lot of time on space, optical technologies, and extreme millimetre-wave engineering in the 70-gigahertz-plus range.

When you came to Mini-Circuits, how did you apply that full-system viewpoint in a component-level business?

That is a great question. My original background was in RF, microwave, and analogue design, after which I attended Stanford for wireless and optical communications, gaining a strong understanding of communication applications. Mini-Circuits operates very much at the component level, whereas some of the roles I held at National Instruments, HP, Facebook, and Amazon were more at the system level. At Mini-Circuits, I have found that